Build Report — Curated Credit Vault on Rayls

From the institutional-blockchain landscape to a working proof of concept

Version 1.1 · July 2026 · Galin Ganchev

> *This is the **implementation deliverable** (~20% of the brief). The **research deliverable** (~80%) — the full platform survey, per-platform L1/L2 architecture, money patterns, capital flows, verification methodology, and the master reference table — is the companion Institutional Blockchain Landscape — Research Report. A non-technical walkthrough of the vault is The Vault in Plain English; the live-demo script is the Demo Runbook — Rayls Credit Vault.*

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1. Executive Summary

The decision: build a curated collateral/liquidity vault for tokenized institutional assets — an ERC-4626 vault over a Morpho-Blue-style lending market — on **Rayls**. USDr deposits earn yield lent against AmFi tokenized-credit collateral, with an issuer-attested NAV oracle (Chainlink-delivered in production, availability on Rayls to be confirmed). The implementation is complete and tested locally (Section 6); deployment fallbacks, in order: Rayls testnet → Robinhood Chain.

Why this and not something else: banks are not asking for another chain — they are asking for compliant settlement, collateral, and asset-servicing workflows that plug into rails they already trust. Of the patterns proven to make independent builders real money (surveyed in full in the companion Institutional Blockchain Landscape — Research Report), the curated vault is the only one that is both institution-shaped and permissionlessly buildable. Rayls is the one venue where "institutional flows actually arriving" and "we can deploy now" intersect: Canton has the flows but is access-gated; Ethereum is open but saturated with curators; Rayls is live, EVM-compatible, has $100M of named collateral landing this month, and runs a $1M+ grant program. Its vault-distribution layer is claimed by a June 2026 partner cohort (Lagoon, Enzyme, Liqvid) — but the lending/borrow-market layer underneath, which such vaults route into on mature chains, remains publicly unclaimed. That is the wedge.

Layer map (three different games — don't confuse them):

- *Middleware* — Chainlink: we consume it, we don't compete with it.
- *Venues* — Rayls, Canton, Robinhood Chain, Ethereum: we deploy on one.
- *Products* — Ondo, Figure, Securitize, this vault: we compete here, and only here.

One-sentence thesis: institutional blockchain value concentrates in settlement, collateral, and compliance workflows — not in owning chains — so we build the first curated collateral vault on Rayls, the one live EVM venue with institutional collateral actually arriving, consuming interoperability middleware rather than competing with it.

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2. Scope & Methodology

Original brief: survey institutional/bank-facing blockchains — Chainlink, Canton, Polkadot, plus every other platform actually used by banks and credit institutions — map their L1/L2 architecture and the financial functionality built on top, identify a dApp use case with a genuine, underserved financial-institutional need, and build a proof-of-concept implementation (~80% research / 20% implementation).

Research method (summary): eight research rounds over ~65 platforms and products in ten categories, plus a verification round that re-checked every load-bearing number against primary sources (correcting eight) and two strategy passes — an adversarial pushback session and a no-deadline resequencing. The full methodology, per-platform findings, sourcing, and the master reference table live in the companion Institutional Blockchain Landscape — Research Report; **this report carries only the conclusions that bear on the build.**

Timeline context: research phase 2026-07-15 → 07-19, implementation 07-20/21, delivery 2026-07-22. The implementation was pulled forward and completed on 07-16 (Section 6).

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3. Why This Use Case (landscape in brief)

The full survey is the companion Institutional Blockchain Landscape — Research Report. Four of its conclusions drive this build:

1. **The institutional flows are owned and closed.** Repo (Broadridge, ~$7–9T/month), interbank payments (JPMorgan Kinexys, ~$7B/day), cross-border messaging (SWIFT + Chainlink CCIP, 11.5k banks), stablecoin reserves (Circle, Tether), securities issuance (Deutsche Börse D7, >€44B) — every large on-chain flow belongs to an incumbent or consortium, and none is open to an outside builder. **Don't compete with rails; attach to one.**
2. **Exactly one money-making pattern is open to a small team today: the curated lending vault.** Of the proven templates — B2B compliance plumbing, fee-on-AUM issuance, own-origination, retail wrappers — only the Morpho-curator vault is both institution-shaped *and* permissionlessly buildable. It already carries real institutional capital (Coinbase routes $1.6B+ through a Steakhouse-curated Morpho vault; Société Générale FORGE runs regulated bank stablecoins through Morpho) and top-tier backing (Morpho raised $175M from a16z/Paradigm, June 2026). Honest expectation: modest curator revenue (~$0.5M ARR even at $1.7B TVL) — a credible *entry* play, not an overnight engine.

3. **Middleware, venues, and products are three different games.** Chainlink is *middleware* — consume it (as the production NAV oracle), don't compete with it. Rayls / Canton / Robinhood Chain / Ethereum are *venues* — deploy on one. Ondo / Figure / Securitize / this vault are *products* — compete here, and only here.
4. **The venue that fits is Rayls.** It is the one place "institutional collateral actually arriving" and "we can deploy now" intersect: live since 2026-04-30, EVM (Solidity/Foundry reuse), USDr gas stablecoin live, **$100M of AmFi tokenized credit arriving July 2026**, a $1M+ developer program — and, crucially, **the lending/borrow-market layer is publicly unclaimed** (the vault-distribution layer already has a June-2026 partner cohort: Lagoon, Enzyme, Liqvid). Canton has the best economics but is Daml- and sponsorship-gated; Ethereum is open but saturated with curators. That open slot on Rayls is the wedge.

One-sentence thesis: *institutional-blockchain value concentrates in settlement, collateral, and compliance workflows — not in owning chains — so we build the first curated collateral vault on Rayls, the one live EVM venue with institutional collateral actually arriving, consuming interoperability middleware rather than competing with it.*

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4. The Venue: Rayls

Architecture — three layers:

1. **Privacy Nodes** — a private EVM chain each institution runs on-prem or in private cloud, for tokenizing and managing assets internally.
2. **Private Networks** — permissioned hub-and-spoke interconnect for institution-to-institution transfers.
3. **Public Chain** — an EVM-compatible public L1, live since 2026-04-30, connecting the private layers to open DeFi.

Built on Reth with periodic state-root anchoring to Ethereum. RBFT consensus, migrating to **Axyl** in 2026 (~250k TPS claimed — internal prototype benchmark, unaudited). Gas is USD-pegged and paid in **USDr**, Rayls' native stablecoin (predictable institutional operating costs). MEV protection and native KYC/KYB identity layers are protocol features. **Enygma** — the privacy protocol (ZK proofs + post-quantum cryptography, selective disclosure for regulators) — reaches the public chain in Q3 2026. Assets move from Privacy Nodes to the Public Chain via a lock-and-mirror bridge (`teleportToPublicChain()` locks on the private side; a relayer mints a mirror token publicly) — likely how AmFi's tokens arrive. Public-chain access is operator-gated by design.

RLS token: 10B fixed supply, no minting. Every transaction fee: 50% burned, 50% to the validator Network Security Pool. TGE 2025-12-01 (~15% circulating; remainder unlocks Dec 2026 → 2029); listed on Coinbase, Gate, MEXC. Distribution: 50% community/ecosystem, 22% investors, 17% contributors, 11% technology provider — non-community allocations on 4-year vesting.

Traction: Núclea (Brazil's largest financial-market infrastructure, 31B transactions/yr) and Cielo (largest card acquirer) live per Rayls; core settlement infrastructure for Brazil's **Drex** CBDC pilot; 25+ institutions claimed. Independently notable: **JPMorgan Kinexys deployed on the network**; **Tether made a strategic investment in Parfin** (Rayls' core developer); **Mastercard added Rayls to its Crypto Partner Program** (March 2026); XP (Brazil's largest retail broker) on the asset side; Parfin's own FX desk migrating ~$400M/month of flows from Q3 2026. Team: ex-BTG Pactual executives plus leads from Siemens, J.P. Morgan DLT, State Street, and AWS; backed by ParaFi, Framework, Valor, and Borderless.

Read critically: the litepaper is marketing material — the TPS figure is an unaudited prototype number and several "live in production" claims are not independently verified. The load-bearing facts above (mainnet date, TGE, the AmFi commitment, Kinexys/Tether/Mastercard) were re-verified against primary sources during the verification round.

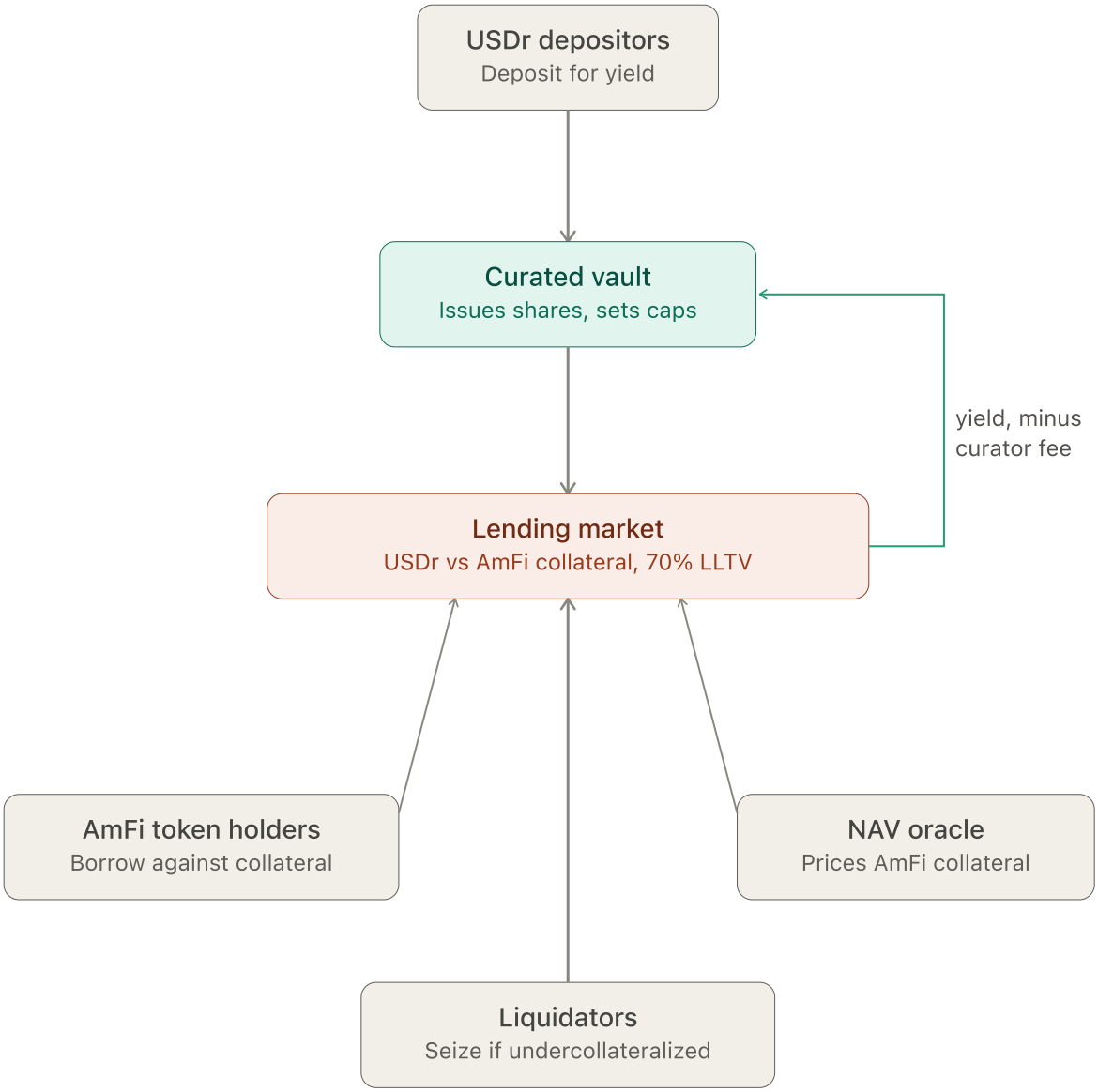
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5. Product Design: A Curated Credit Vault

5.1 The mechanic

A **lending market** is a pool with two assets and one rule. Lenders deposit USDr to earn interest; borrowers lock AmFi tokenized-credit tokens as collateral and borrow USDr against them, overcollateralized (70% liquidation LTV in the PoC). If collateral value falls below the threshold, anyone may permissionlessly **liquidate**: repay the debt and seize collateral at a discount (5% bonus). Why would an institution borrow against tokenized credit instead of selling it? The same reason repo exists in TradFi — cash liquidity without giving up the yield-bearing asset. A holder of receivables yielding ~14% who can borrow at ~8% keeps the spread and redeploys the cash.

A **curated vault** (ERC-4626) sits on top so depositors never touch raw markets: deposit USDr, receive vault shares; the curator decides allocation across markets and caps; depositors get one-click, risk-managed yield; the curator earns a performance fee (15% of yield in the PoC — the rate Steakhouse actually charges on Morpho).



Example: $100 AmFi collateral supports $70 USDr borrowed

Why Rayls specifically, not Ethereum:

1. **Compliance is native, not bolted on.** Rayls has a protocol-level identity layer, so KYC-gated markets don't need a third-party compliance stack — the SG-FORGE-on-Morpho precedent (a regulated bank's stablecoins inside permissionless DeFi) with less friction.
2. **First mover = protocol owner.** On Ethereum this build would be curator #6 (~$0.5M ARR territory). On Rayls nothing like it exists yet — a shot at owning the *protocol* layer, which Morpho's investors valued at ~$2B on a mature chain.
3. **Roadmap stacking.** This vault's liquidity is the cash leg a future repo/collateral-mobility product needs — and repo is the single biggest proven flow in the entire survey ($7–9T/month at Broadridge).

5.2 The collateral: AmFi tokenized credit

AmFi is Brazil's leading private-credit tokenization platform — regulated under the Central Bank of Brazil and the CVM, assets held by licensed custodians, ~R$3B (~$500M+) issued to date. In December 2025, AmFi and

Rayls announced a partnership migrating AmFi's portfolio onto Rayls: **$100M by July 2026 → $500M by January 2027 → $1B+ by June 2027.**

An AmFi token is an on-chain claim on a pool of Brazilian private credit — receivables financing, structured credit notes, FIDC-style senior/junior tranches. Its value tracks the pool's NAV (principal + accrued interest – defaults), and cash flows come from real businesses repaying real loans. That is exactly why it works as collateral (real yield, real NAV) and exactly why it carries real credit risk.

5.3 Risk analysis

- **Oracle.** Tokenized credit has no second-by-second market price. Production needs an issuer-attested NAV feed (Chainlink-delivered if available on Rayls) and conservative LTVs to absorb valuation lag. The implementation enforces a non-zero-NAV invariant so a broken oracle fails loudly instead of silently mispricing collateral.
- **Credit risk does not vanish.** Underlying receivables can sour — Goldfinch lost ~70% of depositor funds (vs. a claimed 20%) on unenforceable credit. Defense: overcollateralization, conservative caps, and never pretending the collateral is risk-free.
- **Liquidation depends on an exit.** Liquidation is permissionless and the discount is the entire incentive — but on Ethereum a liquidator instantly sells seized ETH on a DEX, and AmFi tokens have no equivalent. Three checks, three dead ends: AmFi's own secondary marketplace has said "coming soon" since 2023; no AmFi/USDr pool exists on Rayls' Algebra DEX; AmFi's originator-mediated model makes the tokens buy-and-hold by construction. **The fix — matching how AmFi's real exit mechanics work — is an issuer-redemption pool rather than an open-market sale.** This is built into the implementation, and it doubles as a second product opportunity: solving the redemption gap is what makes RWA collateral usable at all, and nobody else is positioned to claim it either. It does introduce one dependency worth naming: a liquidator holding seized tokens is exposed to the issuer honoring redemptions.
- **Documented, accepted PoC trade-offs** — each recorded as an Architecture Decision Record shipped with the code: the demo concentrates market owner, fee recipient, and NAV setter in a single key (production separates the roles behind a timelock); a single LLTV serves as both borrow limit and liquidation threshold, so a max-borrowed position has zero buffer and becomes liquidatable on the next block's interest (identical to Morpho Blue); and simple interest at spot utilization is marginally gameable — a small repay just before accrual bills the window at a lower rate, a gain that is marginal versus gas.

5.4 Alternatives considered

Concept	Copies	Why not now
Intraday repo / collateral mobility	Broadridge DLR ($7–9T/month), DTCC Collateral AppChain, Canton repo	Needs bank counterparties and months of onboarding — phase 2, with vault liquidity as its cash leg; outreach starts now (Section 7)
LatAm stablecoin/payment orchestration	Stripe/Bridge ($1.1B acquisition), Visa rails ($7B annualized)	Sits on Rayls' own USDr roadmap — joint product or conflict; requires an explicit partner conversation first
Institutional RWA-perps venue	Hyperliquid (~$700–840M/yr revenue, 44% of RWA-perp volume)	Highest ceiling in the survey, but needs deep liquidity and a real risk engine; phase 3 — the lending market's rate curves and NAV outputs are designed to become its price feed

None of these were buildable in the implementation window; all three are roadmap, not rejects.

6. Implementation Status

The vault is built — it is **this repository** (the research document ships with the code it motivated). Built ahead of the nominal window to support the scheduled demo/walkthrough (2026-07-22).

System components (the repository README carries the code-level map):

Component	What it does
Lending market	Morpho-Blue-inspired single market: overcollateralized borrowing (70% liquidation LTV), utilization-based interest rates, permissionless liquidation (5% bonus), virtual-share donation-attack protection, **bad-debt socialization**
Curated vault	ERC-4626 vault over the market: curator-capped allocation, automatic liquidity pull on withdrawals, EIP-4626-compliant max-withdraw/redeem under market illiquidity, 15% performance fee
Mock USDr	Stand-in for Rayls' gas stablecoin (6 decimals)
Mock AmFi credit token	Stand-in collateral: owner-set NAV (oracle stand-in) plus the **issuer-redemption pool** — the Section 5.3 exit fix

Verified end-to-end on a local Anvil chain: a narrated demo script runs deposit → curator allocation → borrow → NAV writedown → liquidation → issuer redemption, with final state independently cross-checked against real on-chain data.

Current state: 77 tests across 5 suites, all passing. Unit and regression tests cover every review finding — **100% branch and statement coverage on all four contracts** — and a stateful **invariant suite** adds seven fitness functions that drive the vault and lending market together — solvency (supplied ≥ borrowed, cash covers advertised liquidity, collateral fully backed, share-sum consistency) plus value conservation (no rounding can mint value on either layer) — held across ~128,000 randomized operation sequences of deposit/withdraw/allocate/borrow/repay/liquidate/fee-change/NAV-move/time-warp. Unit tests prove the paths the author thought of; the invariants prove the ones they didn't. Every load-bearing design choice is additionally recorded as one of **eight Architecture Decision Records** shipped alongside the code (share accounting, bad-debt socialization, the issuer-redemption exit, the NAV trust model, ERC-4626 illiquidity handling, market scope, key management, interest model). **Not yet done:** deployment to the real Rayls testnet — blocked solely on gated network credentials (RPC, chain ID, registry address, operator approval) expected via the partnership, not on anything technical.

7. Action Plan & Roadmap

Steps 1–5 are conversations and checks — cheap, no code, start immediately; 1 and 5 gate the biggest decisions. Step 6 is calendar-gated, so it starts now even though it feels like "later." Steps 7–8 carry the engineering weight.

1. **Ask Rayls the bundled roadmap question** (highest-leverage single action): (a) is a lending/borrow-market layer already privately committed to someone, (b) would LatAm payment orchestration be a joint product or a conflict, (c) what does the $1M developer program's application route require. Same channel: the gated Public Chain credentials — the only thing blocking testnet deployment.

2. **Verify AmFi token mechanics directly with AmFi** — token standard, transferability vs. KYC-gating, on-chain NAV publication, and the *actual* redemption path for a liquidator holding seized tokens. Kills or reshapes the collateral leg; needs a direct answer.
3. **Confirm Rayls deployment specifics** — RPC/chain ID/faucet, permissionless vs. Predicate-gated deployment, and Chainlink oracle availability (else a fallback NAV source is required before any market can price collateral).
4. **Run a technical spike on Zenith** (Solidity-on-Canton) — deploy something trivial and see what breaks; determines whether this codebase has a low-cost second home on Canton.
5. **Open a Canton access conversation** — validator-sponsorship timeline and what "featured app" status (the 62%-pool unlock) actually requires. A relationship gate: the sooner it starts, the sooner Canton is a Year-1 target or ruled out.
6. **Start bank-partner outreach for the repo concept now, not after the vault ships** — its bottleneck is bank-onboarding calendar time, which ticks independently of build progress.
7. **Build the lending market as a real product, not a demo** — multi-market support (AmFi now; Nimofast and deposit tokens as they land), real oracle integration, a proper curator interface, issuer-redemption liquidation tooling — with rate-curve/NAV outputs designed to double as the future perps price feed.
8. **Re-sync checkpoint** once steps 1 and 5 return answers — re-rank everything before new resources move.

Sequencing principle: blockers split into *engineering-time* gates (solved by more build days) and *calendar/relationship* gates (solved only by starting conversations earlier). The vault stays the anchor; repo moves up as a parallel relationship track; payment orchestration folds into the step-1 conversation; perps becomes an architectural constraint on the vault's data outputs. **Run "vault → repo → flow" as overlapping tracks, not a sequence.**

Roadmap:

- **Now:** lending vault on Rayls — proven pattern, working code, grant-eligible.
- **Quarter 1:** compliant lending markets for every asset entering Rayls; bank conversations for repo running in parallel.
- **Year 1:** repo/collateral mobility with the bank pipeline, vault pools as the cash leg.
- **The line:** *"we're building the credit & collateral layer of Rayls."*

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8. References

Sources for this report's build-bearing claims. The full landscape sourcing — all platforms, all categories, the ten-claim verification verdicts — is in the companion Institutional Blockchain Landscape — Research Report and the `Process/` round notes.

The venue — Rayls

- Mainnet (2026-04-30), USDr, ecosystem: Rayls · what's next
- $1M developer program: Rayls
- Rayls × AmFi partnership ($100M → $1B): GlobeNewswire
- Rayls × Lagoon vault cohort: Rayls

The chosen pattern — curated vault

- Morpho $175M round (a16z/Paradigm/Ribbit, ~$2B; $11B+ deposits): Fortune · CoinDesk
- Steakhouse Financial ($1.7B TVL, ~$0.5M ARR, 15% fee): DefiLlama · Morpho

Risk precedents

- Goldfinch wind-down (~$18M defaults, ~70% real losses vs. 20% claimed): The Defiant · DL News
- Figure Q2 2026 ($4.26B quarterly volume, ~$29B cumulative — the "own-origination" alternative): GlobeNewswire

Roadmap alternatives (Section 5.4)

- Broadridge DLR (~$7–9T/month repo — the phase-2 repo target): Jun $7.5T
- Hyperliquid builder codes / RWA-perp volume (the phase-3 perps ceiling): CoinGecko Research

Landscape figures cited in Section 3 (full sourcing in the companion report)

- JPMorgan Kinexys (~$7B/day): DL News
- SWIFT + Chainlink CCIP (production, 11.5k banks): Chainlink Q1 2026 review